

Module2_Lab3

September 13, 2026

Problem 4.4 You are given a list of temperatures (in degrees Celsius) recorded over a week. Write a Python script that categorizes each day's temperature as "Cold", "Warm", or "Hot" based on the following criteria: 1. "Cold" if temperature is below 15 C 2. "Warm" if temperature is between 15 C and 25 C (inclusive) 3. "Hot" if the temperature is above 25 C

```
[2]: Temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```
for temp in temperatures:
    if temp < 15:
        category = "Cold"
    elif 15 <= temp <= 25:
        category = "Warm"
    else:
        category = "Hot"
    print(F"{temp}C: {category}")
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[2], line 2
      1 Temperatures = [10, 16, 23, 28, 14, 22, 30]
----> 2 for temp in temperatures:
      3     if temp < 15:
      4         category = "Cold"
      5     elif 15 <= temp <= 25:

NameError: name 'temperatures' is not defined
```

```
[3]: temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```
for temp in temperatures:
    if temp < 15:
        category = "Cold"
    elif 15 <= temp <= 25:
        category = "Warm"
    else:
        category = "Hot"
    print(f"{temp}C: {category}")
```

```
Cell In[3], line 9
```

```
print(f:{temp}C: {category}")
```

`SyntaxError`: unterminated string literal (detected at line 9)

```
[4]: temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```
for temp in temperatures:
    if temp < 15:
        category = "Cold"
    elif 15 <= 15 temp <= 22:
        category = "Mild"
    elif 23 <= temp <= 26:
        category = "Warm"
    else:
        category = "Hot"

    print(f"{temp}: {category}")
```

Cell In[4], line 6

```
    elif 15 <= 15 temp <= 22:
```

`SyntaxError`: invalid syntax

```
[5]: temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```
for temp in temperatures:
    if temp <15:
        category = "Cold"
    elif 15 <= temp <= 22:
        category = "Mild"
    elif 23 <= temp <= 26:
        category = "Warm"
    else:
        category = "Hot"

    print(f"{temp}: {category}")
```

10: Cold

16: Mild

`TypeError`

Traceback (most recent call last)

Cell In[5], line 9

```
5         category = "Cold"
```

```

6     elif 15 <= temp <= 22:
7         category = "Mild"
8     elif 23 <= temp <= 26:
----> 9         category - "Warm"
10     else:
11         category = "Hot"
12

```

`TypeError: unsupported operand type(s) for -: 'str' and 'str'`

```
[6]: temperatures = [10, 16, 23, 28, 14, 22, 30]
```

```

for temp in temperatures:
    if temp < 15:
        category = "Cold"
    elif 15 <= temp <= 22:
        category = "Mild"
    elif 23 <= temp <= 26:
        category = "Warm"
    else:
        category = "Hot"

    print(f"{temp}: {category}")

```

```

10: Cold
16: Mild
23: Warm
28: Hot
14: Cold
22: Mild
30: Hot

```

I categorized the temperatures to 4 different types: Cold, Mild, Warm, and Hot based on the temperature thresholds provided by the textbook. Used the “elif” for Mild and Warm if temps are between 15 and 22, and 23 and 26, respectively.